

Multiple Choice (10 marks)

1. Find an expression for the coefficient of compressibility at constant temperature for an ideal gas.
 - (a) $(p^2) / (mRT)$
 - (b) $(RT) / (pV)$
 - (c) $(nRT) / (pV^2)$
 - (d) $(mRT) / p^2$
 - (e) $(RT) / p^2$
2. Find the Fourier cosine series over the interval $0 < x < c$ for the function $f(x) = x$.
 - (a) $f(x) \sim (1/2)c + (2c/\pi^2)\sum_{k=1}^{\infty} \frac{1}{(2k)^2} \cos[(2k)x/c]$
 - (b) $f(x) \sim (1/2)c + (4c/\pi^2)\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos[(2k+1)x/c]$
 - (c) $f(x) \sim (1/2)c - (4c/\pi^2)\sum_{n=1}^{\infty} \frac{1}{n^2} \cos[nx/c]$
 - (d) $f(x) \sim (1/2)c - (4c/\pi^2)\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos[(2k+1)x/c]$
 - (e) $f(x) \sim (1/2)c - (4c/\pi^2)\sum_{k=1}^{\infty} \frac{1}{(2k)^2} \cos[(2k)x/c]$
3. Evaluate $f(t)$ if $F(s) = [(2s^2 + 13s + 17) / \{(s+1)(s+3)\}]$
 - (a) $2\delta(t) + (3e^{-t} + e^{-3t})u(t)$
 - (a) $2\delta(t) + (3e^{-t} + 2e^{-3t})u(t)$
 - (a) $3\delta(t) + (3e^{-t} + 2e^{-3t})u(t)$
 - (a) $2 + (3e^{-t} + 2e^{-3t})u(t)$
 - (a) $2 + (5e^{-t} + 2e^{-3t})u(t)$
4. Find the radiation resistance of a single-turn circular loop with a circumference of $(1/4)$ wavelength.
 - (a) 1.27 ohm
 - (b) 1.07 ohm
 - (c) 0.107 ohm
 - (d) 0.77 ohm
 - (e) 0.87 ohm
5. What will be the number of lamps, each having 300 lumens, required to obtain an average illuminance of 50 lux on a 4 m * 3 m rectangular room?

- (a) 6
- (b) 1
- (c) 2
- (d) 5
- (e) 7

True / False (10 marks)

Answer True or False

6. True or False: The inverse Laplace transform of $F(s) = [(2s^2 - 4) / \{(s - 2)(s + 1)(s - 3)\}]$ is $f(t) = [(-4) / 3]e^{-2t} - (1/6)e^{2t} + (7/2)e^{-3t}$.
6. True or False: The vertical frequency in a Lissajous pattern with 5 horizontal and 2 vertical tangencies is 500 Hz when the horizontal frequency is 100 Hz.
7. True or False: The initial current $i(0^+)$ can be expressed as $E/(v(0^+) * R)$ based on the Laplace transform.
7. True or False: The range of the moving coil meter, given the specified parameters, is 2 mA.
8. True or False: The partial-fraction expansion of $F(s)$ is given by $F(s) = [(1/2) / (s + 1)^2] + [(1/4) / (s + 1)] - [(1/4) / (s + 3)]$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the probability of drawing 4 aces consecutively from a standard deck of 52 playing cards without replacement. Analyze the implications of this probability in engineering contexts that involve randomness.
12. Discuss the concept of mean-square quantization error in the context of signal quantization, and derive the expression for this error given that the quantization error is uniformly distributed between $-S/2$ and $+S/2$.

End of Paper